C

IB MATHEMATICS SL • UNIT 5.01

Topic 5: Differential & Integral Calculus

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

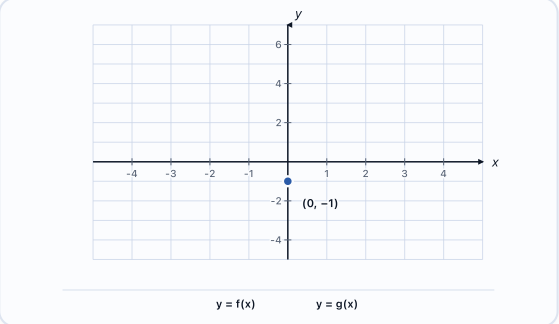
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 B	Q4 A	Q5 B
Q6 B	Q7 A	Q8 A	Q9 D	Q10 A
Q11 D	Q12 C	Q13 A	Q14 C	Q15 C
Q16 A	Q17 C	Q18 A	Q19 D	Q20 D
Q21 B	Q22 D	Q23 C	Q24 D	Q25 B

Topic 5: Differential & Integral Calculus Practice Exercises

1 MCQ

Consider the function $f(x) = x^2 - 5x + 6$. What is the value of $f(2)$?



A 0 ✓

B -2

C 4

D 8

WORKED SOLUTION & VERIFICATION

To find the value of $f(2)$, substitute $x = 2$ into the function $f(x) = x^2 - 5x + 6$.

$f(2) = (2)^2 - 5(2) + 6$

$f(2) = 4 - 10 + 6$

$f(2) = 0$ **[M1] [A1]**

2 MCQ

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

- **A** $y = 2x - 5$ ✓
- **B** $y = -\frac{1}{2}x + 5$
- **C** $y = 2x - 8$
- **D** $y = -2x + 11$

WORKED SOLUTION & VERIFICATION

In a Voronoi diagram, the boundary edge between two sites is along the perpendicular bisector of the line segment connecting them.

1. Find the midpoint M of the segment AB:

$$M = (\frac{2+6}{2}, \frac{4+2}{2}) = (4, 3)$$

2. Find the gradient of segment AB:

$$m_{AB} = \frac{2-4}{6-2} = \frac{-2}{4} = -\frac{1}{2}$$

3. The gradient of the perpendicular bisector is the negative reciprocal of m_{AB}:

$$m_{\perp} = -\frac{1}{-\frac{1}{2}} = 2$$

4. Use the point-slope form with midpoint $(4, 3)$ and gradient $m_{\perp} = 2$:

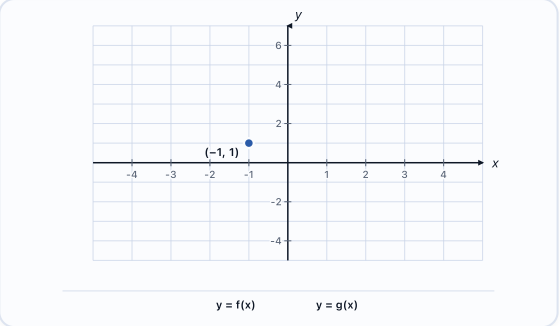
$$y - 3 = 2(x - 4) \implies y = 2x - 8 + 3 \implies y = 2x - 5$$

Common Misconceptions:

- Choice B uses the gradient of AB rather than its negative reciprocal.
- Choice C correctly finds the gradient $m = 2$ but forgets to add the y-coordinate of the midpoint, setting $y = 2(x - 4) = 2x - 8$.
- Choice D uses $m = -2$ instead of $+2$. **[M1] [A1]**

3 MCQ

Consider the function $f(x) = x^2 - 2x + 1$. Determine the y-coordinate of the vertex of the graph of $y = f(x)$.



A 1

B 0 ✓

C -1

D 2

WORKED SOLUTION & VERIFICATION

The graph of a quadratic function $f(x) = ax^2 + bx + c$ is a parabola. The x-coordinate of the vertex is given by $x = -\frac{b}{2a}$. For the function $f(x) = x^2 - 2x + 1$, we have $a = 1$ and $b = -2$. Therefore, the x-coordinate of the vertex is $x = -\frac{-2}{2(1)} = \frac{2}{2} = 1$. To find the y-coordinate of the vertex, we substitute this value of x back into the function: $f(1) = (1)^2 - 2(1) + 1 = 1 - 2 + 1 = 0$. Thus, the y-coordinate of the vertex is 0. **[M1] [A1]**

4 MCQ

Consider the function $f(x) = 3x^2 - 2x + 5$. Find the value of $f'(1)$.

A 4 ✓

B 5

C 6

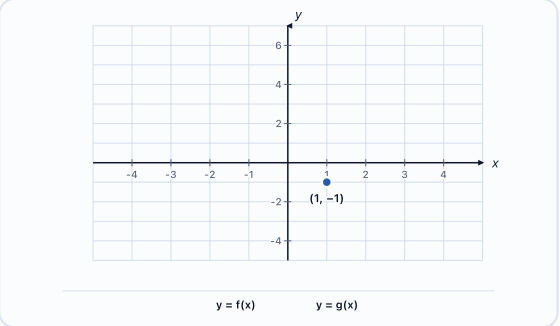
D 7

WORKED SOLUTION & VERIFICATION

The derivative of a function $f(x)$ gives the instantaneous rate of change of the function at any given point. We first find the derivative of $f(x) = 3x^2 - 2x + 5$ using the power rule: $f'(x) = \frac{d}{dx}(3x^2) - \frac{d}{dx}(2x) + \frac{d}{dx}(5)$. Applying the power rule, $\frac{d}{dx}(ax^n) = nax^{n-1}$ and the derivative of a constant is 0, we get $f'(x) = 2(3x^{2-1}) - 1(2x^{1-1}) + 0$, which simplifies to $f'(x) = 6x - 2$. To find the value of the derivative at $x = 1$, we substitute $x = 1$ into the expression for $f'(x)$: $f'(1) = 6(1) - 2 = 6 - 2 = 4$. **[M1] [A1]**

5 MCQ

The graph of the function $f(x) = 2x^2 - 3x + 5$ is shown below. Determine the value of $f'(2)$.



A 7

B 5 ✓

C 9

D 2

WORKED SOLUTION & VERIFICATION

The derivative of the function $f(x) = 2x^2 - 3x + 5$ is found using the power rule: $f'(x) = \frac{d}{dx}(2x^2) - \frac{d}{dx}(3x) + \frac{d}{dx}(5)$. This gives $f'(x) = 4x - 3$. To find $f'(2)$, substitute $x = 2$ into the derivative: $f'(2) = 4(2) - 3 = 8 - 3 = 5$. **[M1] [A1]**

6 MCQ

Consider the function $f(x) = x^2 - 6x + 5$. Find the x-coordinate of the vertex of the parabola represented by $y = f(x)$.

A 1

B 3 ✓

C -1

D -3

WORKED SOLUTION & VERIFICATION

The given function is a quadratic function in the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -6$, and $c = 5$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values of a and b, we get $x = -\frac{-6}{2(1)} = \frac{6}{2} = 3$. Therefore, the x-coordinate of the vertex is 3. **[M1] [A1]**

7 MCQ

Find $\int x^2 + 4x^3\, dx$.

A $\frac{1}{3}x^3 + x^4 + C$ ✓

B $\frac{1}{4}x^4 + \frac{4}{5}x^5 + C$

C $3x^3 + 16x^4 + C$

D $x^2 + 4x^3 + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\, dx = \frac{x^{n+1}}{n+1} + C$: $\int x^2 + 4x^3\, dx = \frac{1}{3}x^3 + x^4 + C$. **[M1] [A1]**

8 MCQ

Find $\int 5x^2 + 4x\,dx$.

- **A** $\frac{5}{3}x^3 + 2x^2 + C$ ✓
- **B** $\frac{5}{4}x^4 + \frac{4}{3}x^3 + C$
- **C** $15x^3 + 8x^2 + C$
- **D** $5x^2 + 4x + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int 5x^2 + 4x\,dx = \frac{5}{3}x^3 + 2x^2 + C$. **[M1] [A1]**

9 MCQ

Find $f'(x)$ given $f(x) = x^4 + 4x^2 + 2x$.

- **A** $5x^4 + 12x^2 + 4x$
- **B** $x^4 + 4x^2 + 2x$
- **C** $x^3 + 4x + 2$
- **D** $4x^3 + 8x + 2$ ✓

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Differentiate term by term using the power rule: $f'(x) = 4x^3 + 8x + 2$. **[M1] [A1]**

10 MCQ

Find $\int -x^3 - 2\,dx$.

- **A** $-\frac{1}{4}x^4 - 2x + C$ ✓
- **B** $-x^3 - 2 + C$
- **C** $-\frac{1}{5}x^5 - x^2 + C$
- **D** $-4x^4 - 2x + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int -x^3 - 2\,dx = -\frac{1}{4}x^4 - 2x + C$. **[M1] [A1]**

11 MCQ

Find $f'(x)$ given $f(x) = 6x^4 - 4x^3 - 2x - 5$.

- **A** $6x^4 - 4x^3 - 2x$
- **B** $30x^4 - 16x^3 - 4x$
- **C** $6x^3 - 4x^2 - 2$
- **D** $24x^3 - 12x^2 - 2$ ✓

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Differentiate term by term using the power rule: $f'(x) = 24x^3 - 12x^2 - 2$. **[M1] [A1]**

12 MCQ

Find $\int\ 3+4x^3\,dx$.

A $3+4x^3+C$

B $3x+16x^4+C$

C $3x+x^4+C$ ✓

D $\frac{3}{2}x^2+\frac{4}{5}x^5+C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx=\frac{x^{n+1}}{n+1}+C$: $\int 3+4x^3\,dx=3x+x^4+C$. **[M1]**

[A1]

13 MCQ

Calculate $\int\ 5x^3+2x\,dx$.

A $\frac{5}{4}x^4+x^2+C$ ✓

B $x^5+\frac{2}{3}x^3+C$

C $20x^4+4x^2+C$

D $5x^3+2x+C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx=\frac{x^{n+1}}{n+1}+C$: $\int 5x^3+2x\,dx=\frac{5}{4}x^4+x^2+C$. **[M1]**

[A1]

14 MCQ

Find $\int\ 6-2x^2\,dx$.

A $6x-6x^3+C$

B $6-2x^2+C$

C $6x-\frac{2}{3}x^3+C$ ✓

D $3x^2-\frac{1}{2}x^4+C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx=\frac{x^{n+1}}{n+1}+C$: $\int 6-2x^2\,dx=6x-\frac{2}{3}x^3+C$. **[M1]**

[A1]

15 MCQ

Find $\int -4x^2 - 3\,dx$.

A $-12x^3 - 3x + C$

B $-x^4 - \frac{3}{2}x^2 + C$

C $-\frac{4}{3}x^3 - 3x + C$ ✓

D $-4x^2 - 3 + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int -4x^2 - 3\,dx = -\frac{4}{3}x^3 - 3x + C$.

[M1] [A1]

16 MCQ

Find $\int -5x + 3x^3\,dx$.

A $-\frac{5}{2}x^2 + \frac{3}{4}x^4 + C$ ✓

B $-\frac{5}{3}x^3 + \frac{3}{5}x^5 + C$

C $-10x^2 + 12x^4 + C$

D $-5x + 3x^3 + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int -5x + 3x^3\,dx = -\frac{5}{2}x^2 + \frac{3}{4}x^4 + C$.

[M1] [A1]

17 MCQ

Find $\int -2 + 3x\,dx$.

A -2+3x+C

B $-2x + 6x^2 + C$

C $-2x + \frac{3}{2}x^2 + C$ ✓

D $-x^2 + x^3 + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int -2 + 3x\,dx = -2x + \frac{3}{2}x^2 + C$. **[M1]**

[A1]

18 MCQ

Find $\int -2x^2 + 5\,dx$.

- **A** $-\frac{2}{3}x^3 + 5x + C$ ✓
- **B** $-\frac{1}{2}x^4 + \frac{5}{2}x^2 + C$
- **C** $-2x^2 + 5 + C$
- **D** $-6x^3 + 5x + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int -2x^2 + 5\,dx = -\frac{2}{3}x^3 + 5x + C$. **[M1] [A1]**

19 MCQ

Find $f'(x)$ given $f(x) = 2x^3 + x^2 - 4$.

- **A** $2x^3 + x^2$
- **B** $2x^2 + x$
- **C** $8x^3 + 3x^2$
- **D** $6x^2 + 2x$ ✓

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Differentiate term by term using the power rule: $f'(x) = 6x^2 + 2x$. **[M1] [A1]**

20 MCQ

Find $\int x - 4x^3\,dx$.

- **A** $x - 4x^3 + C$
- **B** $2x^2 - 16x^4 + C$
- **C** $\frac{1}{3}x^3 - \frac{4}{5}x^5 + C$
- **D** $\frac{1}{2}x^2 - x^4 + C$ ✓

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int x - 4x^3\,dx = \frac{1}{2}x^2 - x^4 + C$. **[M1] [A1]**

21 MCQ

Calculate $\int -3x^3 - 1\,dx$.

- **A** $-\frac{3}{5}x^5 - \frac{1}{2}x^2 + C$
- **B** $-\frac{3}{4}x^4 - x + C$ ✓
- **C** $-3x^3 - 1 + C$
- **D** $-12x^4 - x + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int -3x^3 - 1\,dx = -\frac{3}{4}x^4 - x + C$. **[M1] [A1]**

22 MCQ

Find $f'(x)$ given $f(x) = 5x^4 + 4x^3 - 5x$.

A $25x^4 + 16x^3 - 10x$

B $5x^3 + 4x^2 - 5$

C $5x^4 + 4x^3 - 5x$

D $20x^3 + 12x^2 - 5$ ✓

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Differentiate term by term using the power rule: $f'(x) = 20x^3 + 12x^2 - 5$. **[M1] [A1]**

23 MCQ

Find $f'(x)$ given $f(x) = 4x^4 + 5x^2 + 4x + 1$.

A $4x^4 + 5x^2 + 4x$

B $20x^4 + 15x^2 + 8x$

C $16x^3 + 10x + 4$ ✓

D $4x^3 + 5x + 4$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Differentiate term by term using the power rule: $f'(x) = 16x^3 + 10x + 4$. **[M1] [A1]**

24 MCQ

Find $\int -x - 3x^3\,dx$.

A $-\frac{1}{3}x^3 - \frac{3}{5}x^5 + C$

B $-x - 3x^3 + C$

C $-2x^2 - 12x^4 + C$

D $-\frac{1}{2}x^2 - \frac{3}{4}x^4 + C$ ✓

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int -x - 3x^3\,dx = -\frac{1}{2}x^2 - \frac{3}{4}x^4 + C$. **[M1] [A1]**

25 MCQ

Find $f'(x)$ given $f(x) = 3x^4 - 3x^2 - 4x$.

A $3x^4 - 3x^2 - 4x$

B $12x^3 - 6x - 4$ ✓

C $15x^4 - 9x^2 - 8x$

D $3x^3 - 3x - 4$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Differentiate term by term using the power rule: $f'(x) = 12x^3 - 6x - 4$. **[M1] [A1]**